

Pratyansha Rana

Software Engineer

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[Portfolio](#) | [Github](#) | [LinkedIn](#)

Professional Summary

Passionate Software Engineer focused on building robust, high-integrity systems. Fast learner with a commitment to technical excellence and system scalability rather than superficial feature sets.

Technical Skills

- **Frontend & Mobile:** React Native, Figma
- **Backend & Database:** Express.js, MongoDB, Firebase, Supabase
- **Cloud & Tools:** Google Cloud Platform (GCP), Git, Project Management
- **Specialized:** Quantum Computing, Quantum Key Distribution (QKD)

Achievements

- **Smart India Hackathon 2025:** Led a 6-member team for Problem Statement SIH25031; reached the national-level waitlist stage through coordinated idea validation and submission.
- **1st Runner-Up - InnoMaker Project Showcase:** Achieved 2nd place and ₹10,000 prize at SGSU Bhopal (March 2026) for "QUBES," focusing on practical technical implementation.

Projects

- **Phantom:** Secure closed-group defense communication platform utilizing Hybrid Post-Quantum Cryptography (Kyber).
 - [GitHub Link](#)
- **LokAwaz:** Civic issue reporting platform with real-time pothole detection via camera streams and automated staff assignment. Built with TypeScript and React Native.
 - [GitHub Link](#)
- **QUBES:** Quantum Key Distribution (QKD) simulation integrated into a secure chat application.
 - [Live Demo](#)
- **GearSwap:** Dedicated in-campus marketplace application for peer-to-peer student transactions.
 - [Live Demo](#)
- **Modified-BB84 (Research):** Python-based simulation modifying the BB84 QKD protocol for Topological Quantum Systems.
 - [GitHub Link](#)

Education

LNCT&S (Bhopal) | B-Tech CSE (AI & ML) | Expected 2028 (Current 4th Sem)